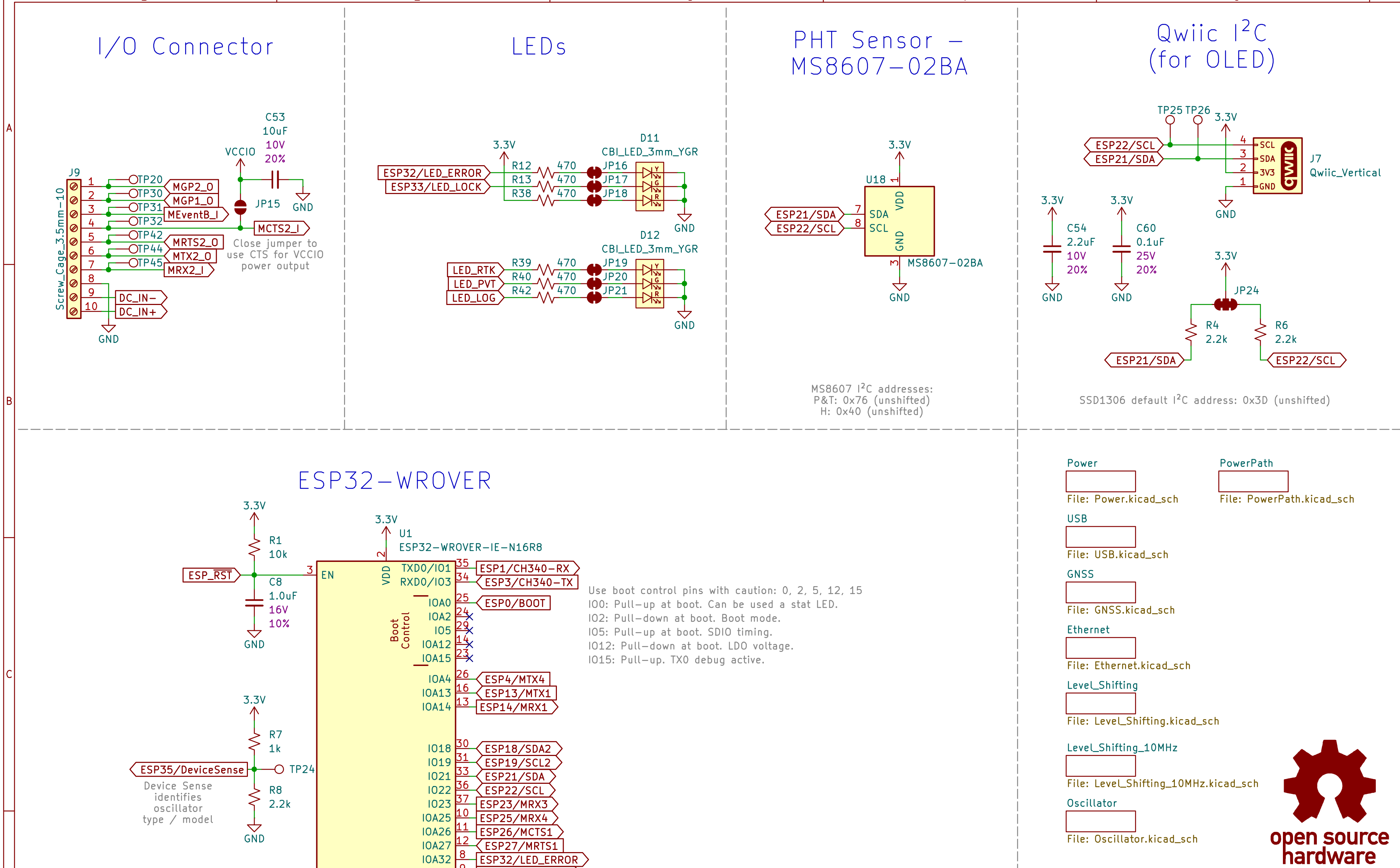


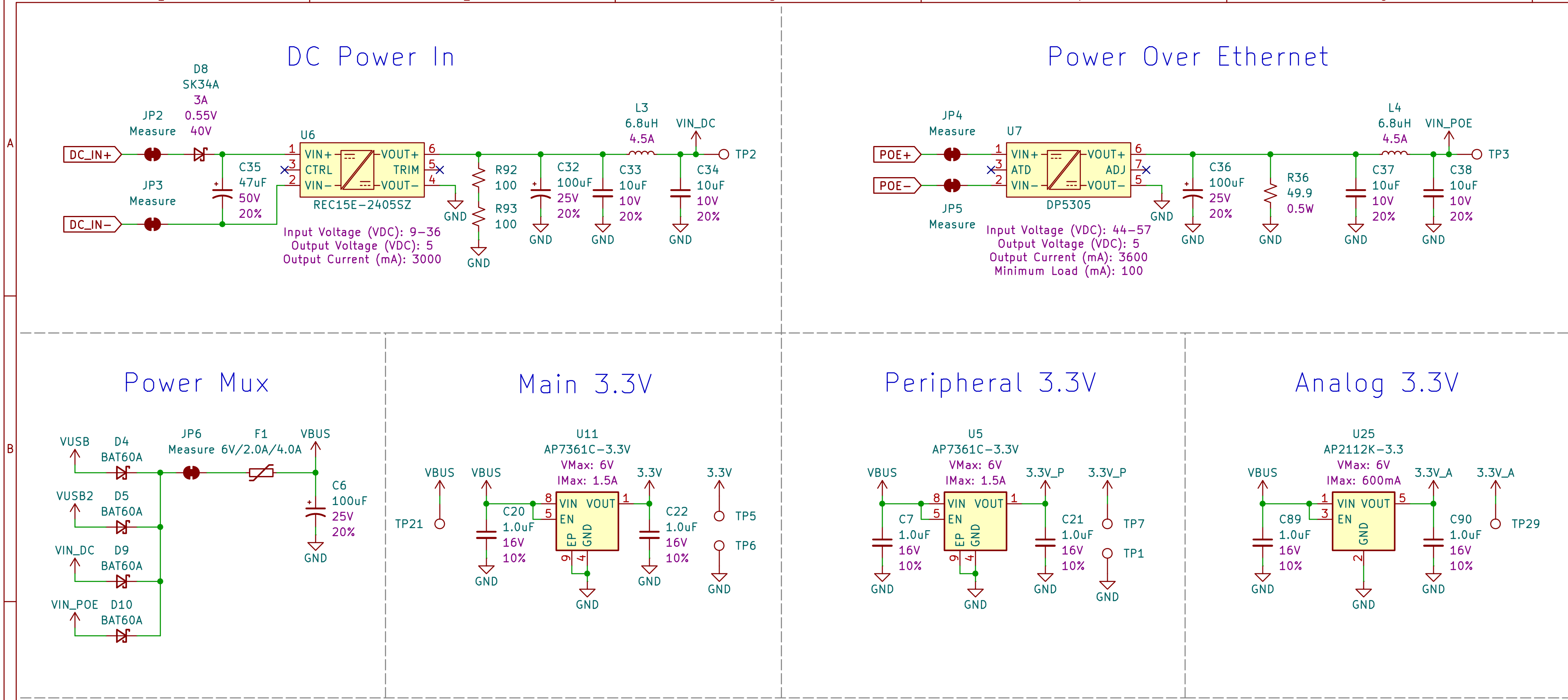
SPARKPNT

Designed by: P.C.
SparkFun Electronics

Sheet: /
File: SparkPNT_GNSSDO_Plus.kicad_sch

Title: GNSSDO Plus (mosaic-T, STP3593LF)

Size: USLetter	Date: 2025-06-06	**Rev: v02**
KiCad E.D.A. 9.0.1		Id: 1/9



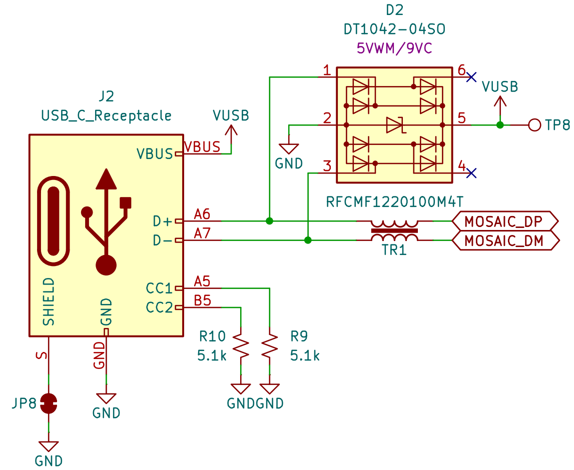
Sheet: /Power/
File: Power.kicad_sch

Title: GNSSDO Plus – Power

Size: USLetter | Date: | Rev:

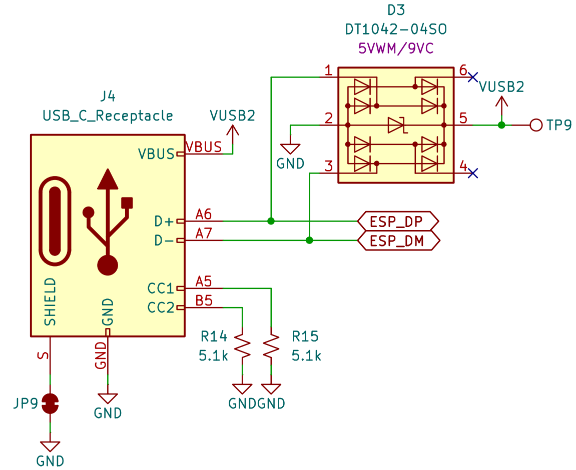
KiCad E.D.A. 9.0.1 | Id: 2/9

Mosaic USB

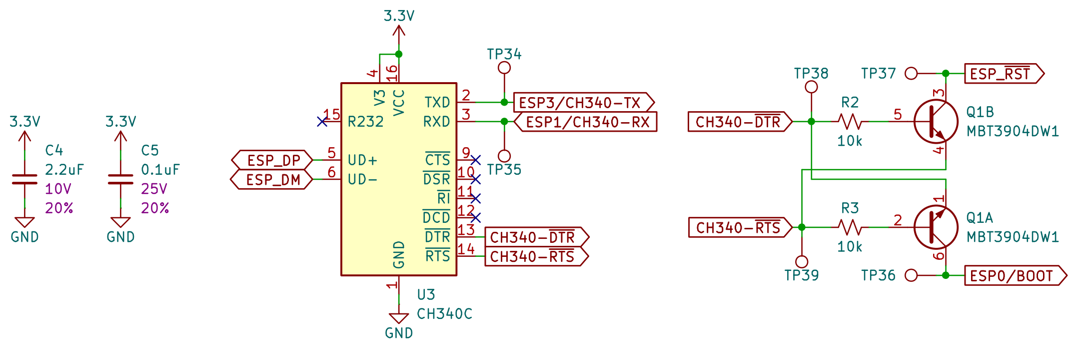


USB Track Impedance: Differential Pair
https://saturnpcb.com/saturn-pcb-toolkit/
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
10.5 mil track with 9.5 mil gap (20 mil center to center) = 90 Ohms

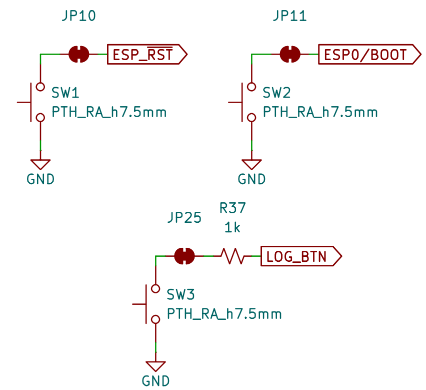
ESP32 USB



ESP32 USB to Serial – CH340C



Buttons

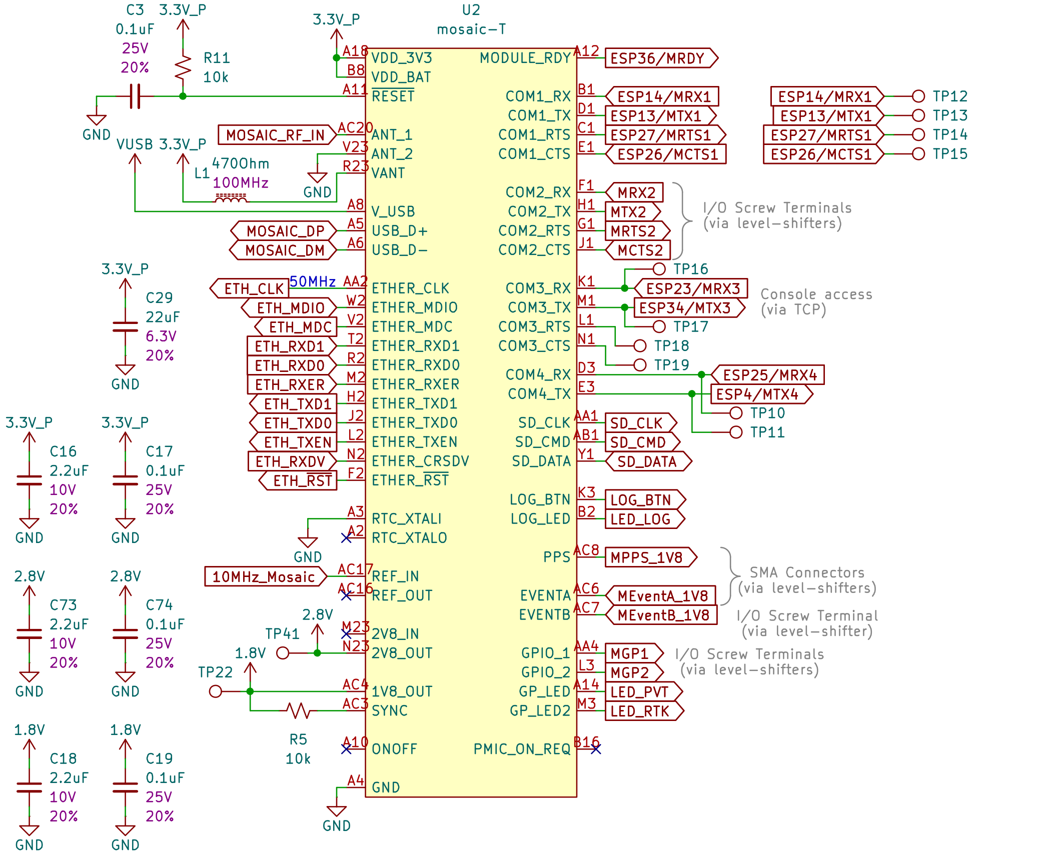


Sheet: /USB/
File: USB.kicad_sch

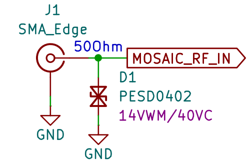
Title: GNSSDO Plus – USB

Size: USLetter	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 3/9

mosaic Tri-band GNSS

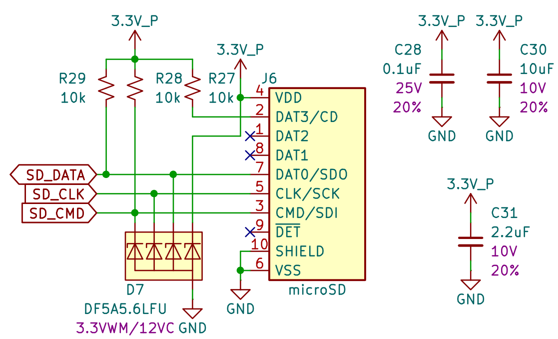


GNSS Antenna



Microstrip Calculation:
Copper Thickness (1oz): 1.4mil/0.035mm
Board thickness: 1.6mm
Dielectric thickness (layer 1 to 2): 0.2mm
Er: 4.6
Polygon Isolation: 6mil/0.1524mm
RF Trace Width: 13mil/0.33mm
https://chemandy.com/calculators/coplanar-waveguide-with-ground-calculator.htm

microSD

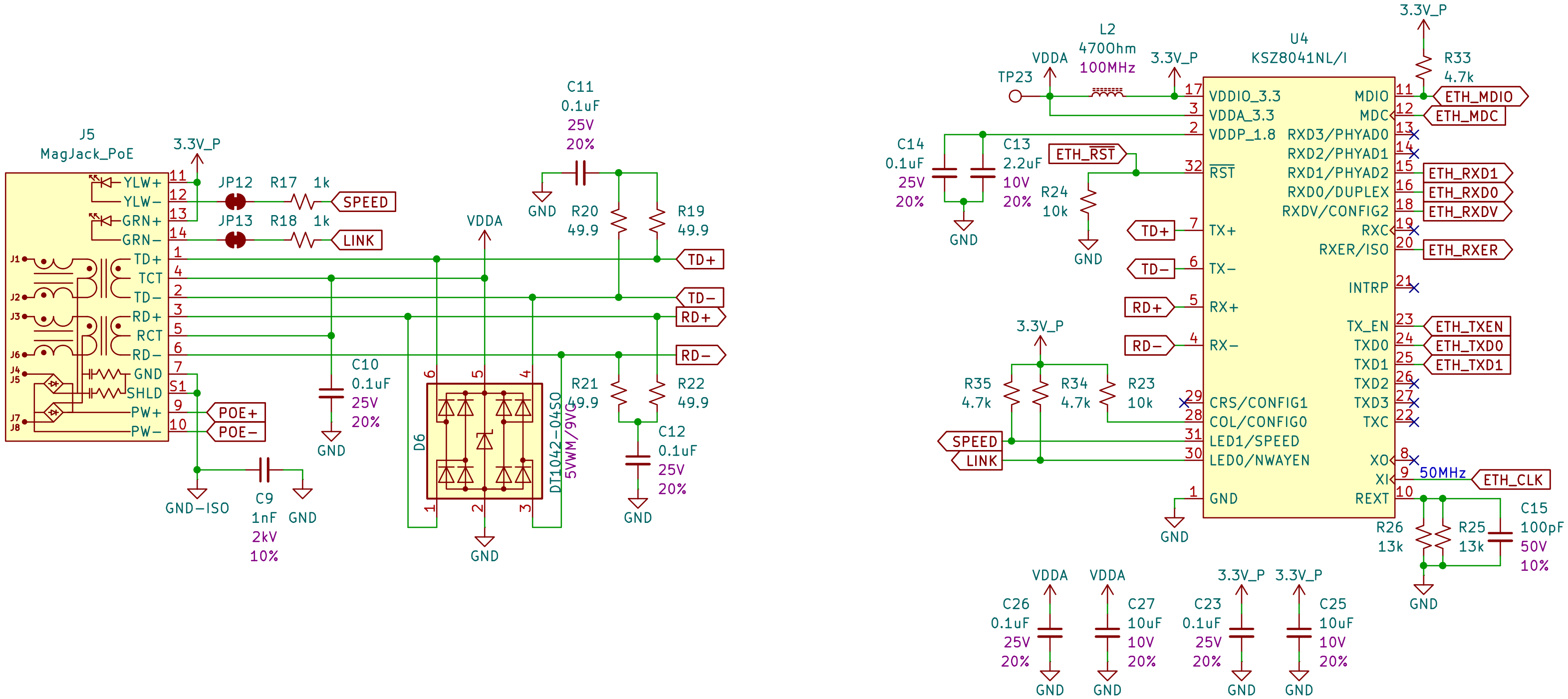


Sheet: /GNSS/
File: GNSS.kicad_sch

Title: GNSSDO Plus – GNSS

Size: USLetter	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 4/9

Ethernet

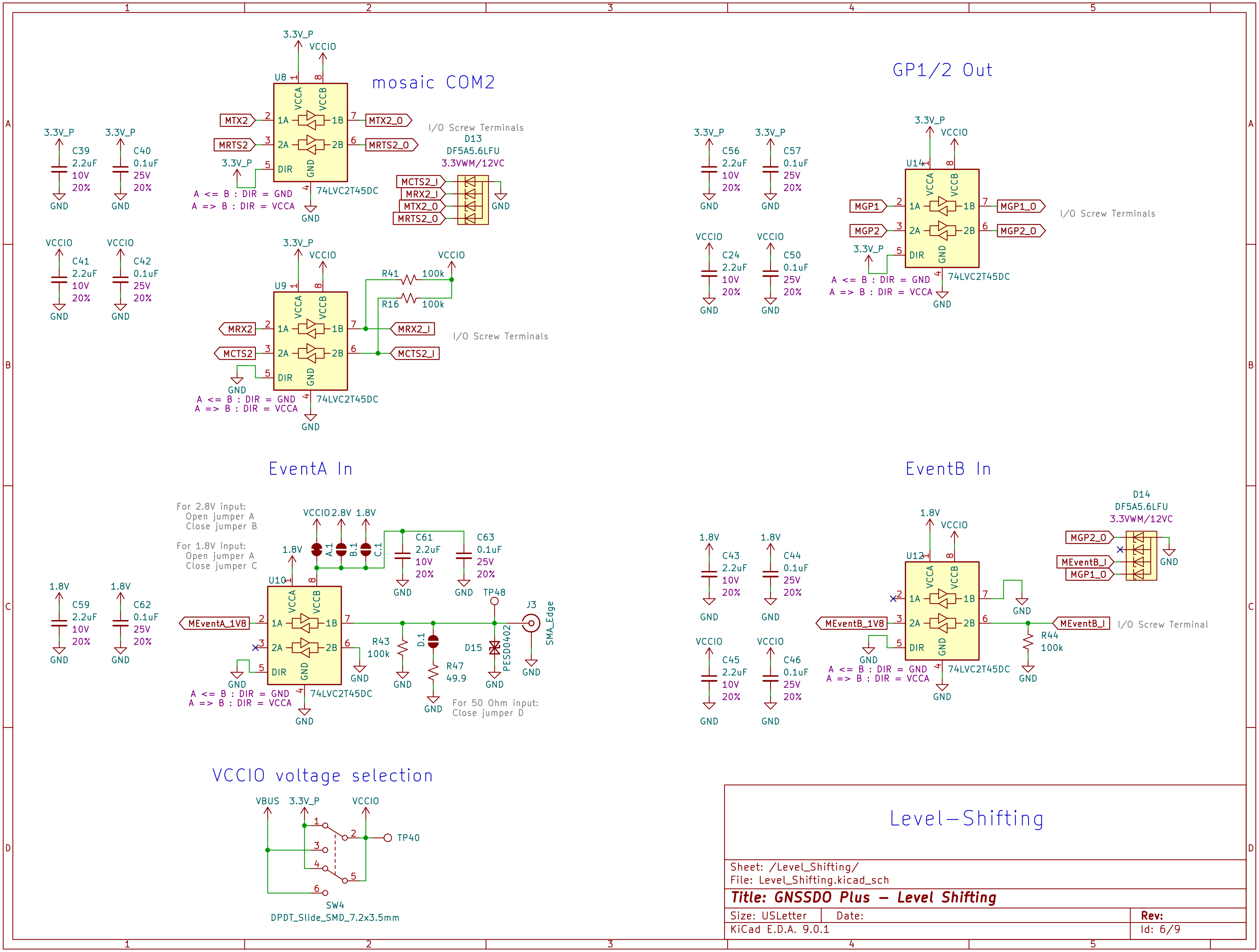


Ethernet Track Impedance: Differential Pair
https://saturnpcb.com/saturn-pcb-toolkit/
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
9.0 mil track with 11.0 mil gap (20 mil center to center) = 100 Ohms
Each pair should match in length to better than 0.5mm

Sheet: /Ethernet/
File: Ethernet.kicad_sch

Title: GNSSDO Plus − Ethernet

Size: USLetter	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 5/9

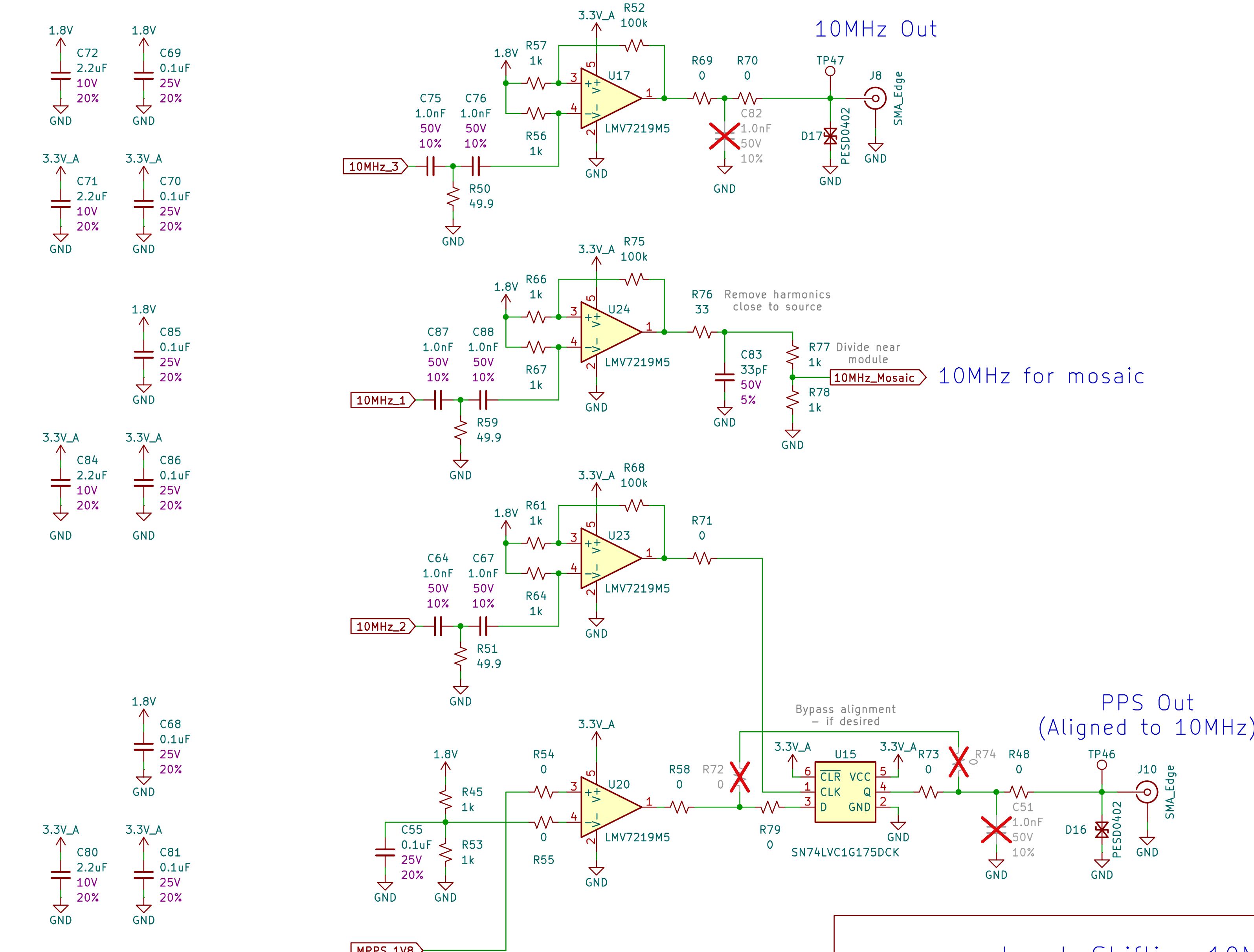


Level-Shifting

Sheet: /Level_Shifting/
File: Level_Shifting.kicad_sch

Title: GNSSDO Plus – Level Shifting

Size: USLetter	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 6/9



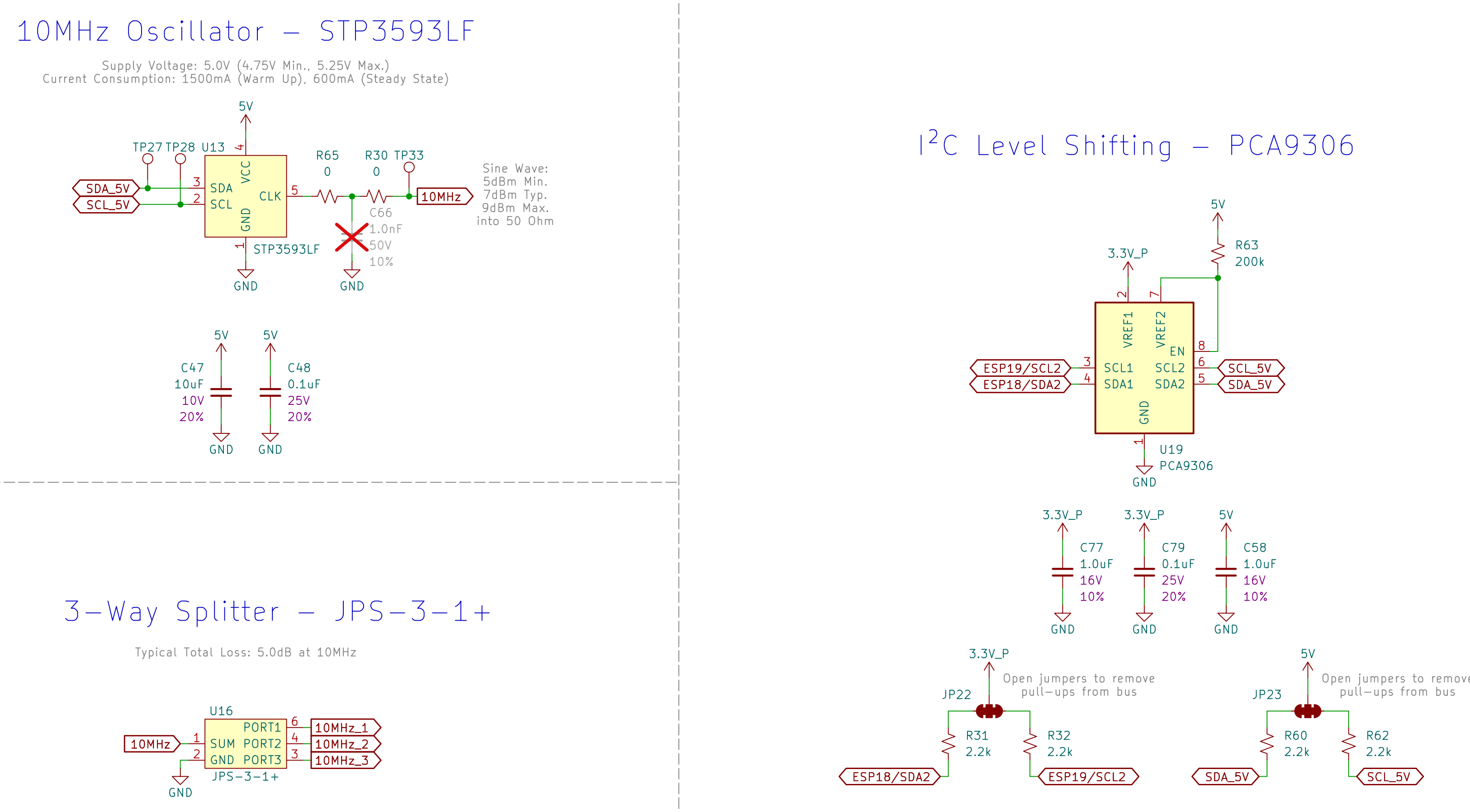
Note: all paths use identical components to ensure equal temperature dependence

Level–Shifting 10MHz

Sheet: /Level_Shifting_10MHz/
File: Level_Shifting_10MHz.kicad_sch

Title: GNSSDO Plus – Level Shifting 10MHz

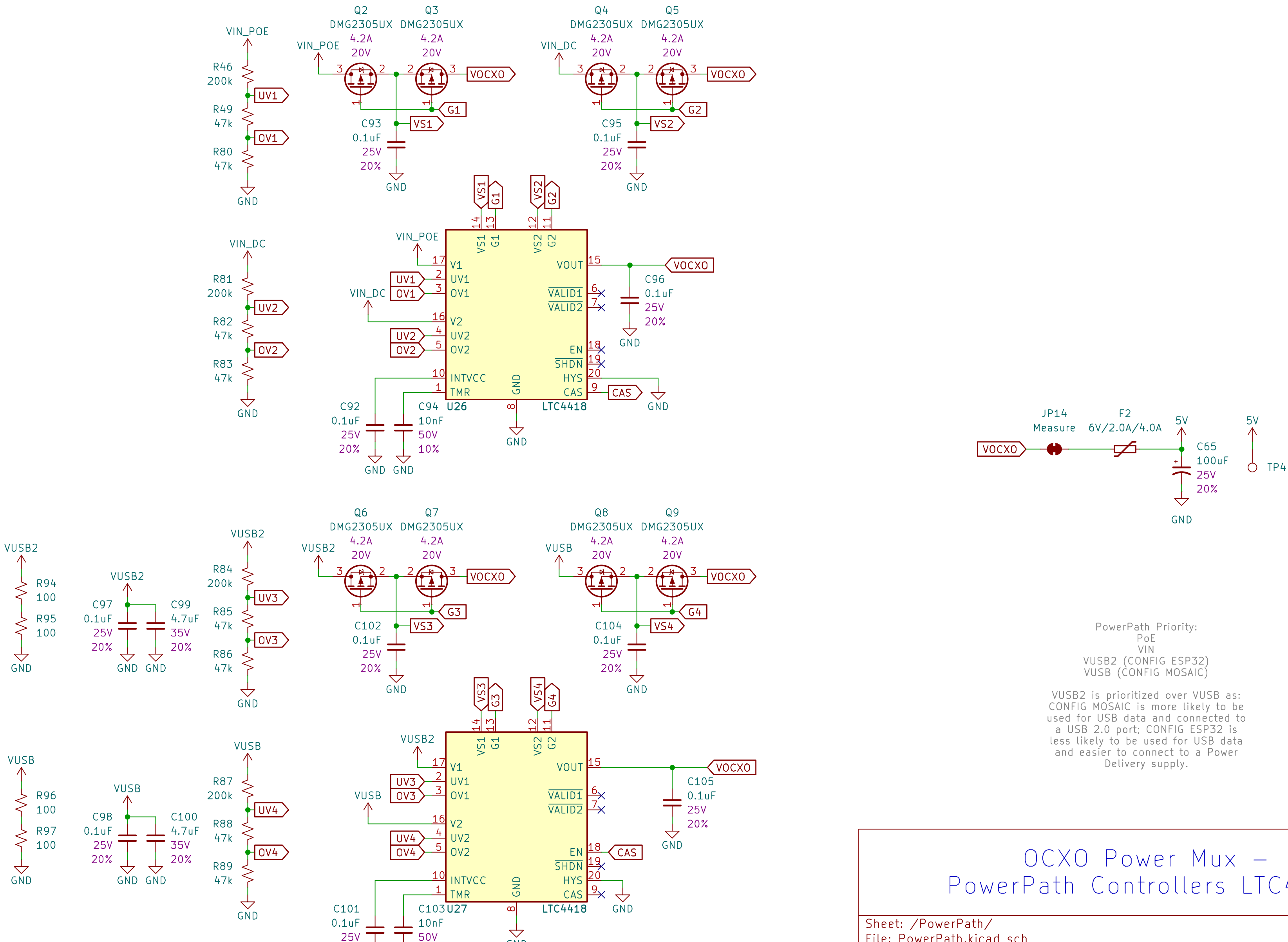
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 9.0.1 | Id: 7/9



Sheet: /Oscillator/
File: Oscillator.kicad_sch

Title: GNSSDO Plus – Oscillator

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 8/9



PowerPath Priority:
PoE
VIN
VUSB2 (CONFIG ESP32)
VUSB (CONFIG MOSAIC)

VUSB2 is prioritized over VUSB as: CONFIG MOSAIC is more likely to be used for USB data and connected to a USB 2.0 port; CONFIG ESP32 is less likely to be used for USB data and easier to connect to a Power Delivery supply.

Resistor network calculated using:
http://beta-tools.analog.com/ltpowertools/LTC4418/
UnderVoltage: 3.2V
OverVoltage: 6.2V

OCXO Power Mux – PowerPath Controllers LTC4418

Sheet: /PowerPath/
File: PowerPath.kicad_sch

Title: GNSSDO Plus – OCXO Power Mux (PowerPath)

Size: A4 | Date: | Rev:
KiCad E.D.A. 9.0.1 | Id: 9/9